

$$\Rightarrow E_{\lambda_j} \cap E_{\lambda_{j-1}} + \dots + E_{\lambda_1} = \{\vec{0}\}$$

Definition: A linear transformation $T: V \rightarrow V$ is called diagonalizable if there exists an ordered basis B of V such that $[T]_B$ is a diagonal matrix.

Theorem. Let V be an n -dimensional vector space. Let $T: V \rightarrow V$ be a linear transformation. Let $\lambda_1, \dots, \lambda_k$ be all the distinct eigenvalues of T . Let B_i be an ordered basis for E_{λ_i} , for all $i=1, 2, \dots, k$. Then the following statements are equivalent.

1). T is diagonalizable

2). The characteristic polynomial of T is

$$f(\lambda) = (\lambda - \lambda_1)^{d_1} (\lambda - \lambda_2)^{d_2} \dots (\lambda - \lambda_k)^{d_k}$$

with $d_1 + \dots + d_k = n$ and $d_1 = \dim E_{\lambda_1}$, $d_2 = \dim E_{\lambda_2}$, $\dots$, $d_k = \dim E_{\lambda_k}$.

3). $\dim(E_{\lambda_1}) + \dim(E_{\lambda_2}) + \dots + \dim(E_{\lambda_k}) = n$

4). $B_1 \cup B_2 \cup \dots \cup B_k$ forms a basis for V .

(5). the minimal polynomial of T has no repeated roots.

end of Feb 10

Proof: (1) $\Rightarrow$ (2): Suppose T is diagonalizable.

$\Rightarrow$ there exists a basis $B = \{\vec{v}_1, \dots, \vec{v}_n\}$ such that

$$[T]_B = \begin{pmatrix} \alpha_1 & & \\ & \ddots & \\ & & \alpha_n \end{pmatrix}$$

$\Rightarrow f(\lambda) = \det(T - \lambda I) = (\alpha_1 - \lambda) \dots (\alpha_n - \lambda) \Rightarrow \alpha_1, \alpha_2, \dots, \alpha_n$ are all the eigenvalues

of T . (note $f(x)$ is factored into all linear factors).

As $\lambda_1, \dots, \lambda_k$ are all the distinct eigenvalues of T

$$\Rightarrow (\alpha_1 - \lambda)(\alpha_2 - \lambda) \dots (\alpha_n - \lambda) = (\lambda_1 - \lambda)^{d_1} (\lambda_2 - \lambda)^{d_2} \dots (\lambda_k - \lambda)^{d_k} \quad \text{with } d_1 + d_2 + \dots + d_k = n.$$

WLOG, we may rearrange the vectors in B such that

$$[T]_B = \begin{pmatrix} \lambda_1 & & & \\ & \ddots & & \\ & & \lambda_2 & \\ & & & \ddots \\ & & & & \lambda_k & \\ & & & & & \ddots \\ & & & & & & \lambda_k \end{pmatrix}$$

$\uparrow$ d_1 copies $\uparrow$ d_2 copies $\uparrow$ d_k copies.

$$\Rightarrow T(\vec{v}_1) = \lambda_1 \vec{v}_1, \dots, T(\vec{v}_{d_1}) = \lambda_1 \vec{v}_{d_1}; \quad T(\vec{v}_{d_1+1}) = \lambda_2 \vec{v}_{d_1+1}, \dots, T(\vec{v}_{d_1+d_2}) = \lambda_2 \vec{v}_{d_1+d_2}, \dots$$

$$\Rightarrow \{\vec{v}_1, \dots, \vec{v}_{d_1}\} \subseteq E_{\lambda_1}, \quad \{\vec{v}_{d_1+1}, \dots, \vec{v}_{d_1+d_2}\} \subseteq E_{\lambda_2}, \quad \dots \quad \{\vec{v}_{d_1+\dots+d_{k-1}+1}, \dots, \vec{v}_n\} \subseteq E_{\lambda_k}$$

As $\{\vec{v}_1, \dots, \vec{v}_n\}$ is linearly independent

$$\dim(E_{\lambda_1}) \geq d_1, \quad \dim(E_{\lambda_2}) \geq d_2, \quad \dots, \quad \dim(E_{\lambda_k}) \geq d_k.$$

$\downarrow$ "=" if and only if $\dim(E_{\lambda_i}) = d_i$ for all $i=1, 2, \dots, k$

$$n = d_1 + d_2 + \dots + d_k \leq \dim(E_{\lambda_1}) + \dots + \dim(E_{\lambda_k}) = \dim(E_{\lambda_1} \oplus E_{\lambda_2} \oplus \dots \oplus E_{\lambda_k}) \leq \dim(F^n) = n$$

$$\Rightarrow \dim(E_{\lambda_1} \oplus \dots \oplus E_{\lambda_k}) = n \quad \text{and} \quad \dim(E_{\lambda_i}) = d_i \quad \text{for all } i=1, \dots, k.$$

(2) $\Rightarrow$ (3): obvious.

(3) $\Rightarrow$ (4): As $B_1 \cup \dots \cup B_k$ contains linearly independent vectors and there are exactly

n vectors, $B_1 \cup \dots \cup B_k$ is a basis for V .

(4) $\Rightarrow$ (1). For any vector $\vec{v} \in B$: $T(\vec{v}) = \lambda_i \vec{v}$

$\Rightarrow [T]_B$ is a diagonal matrix. $\square$

Theorem: (Schur Factorization). Let $A \in \mathbb{C}^{n \times n}$. Then A is similar to an upper triangular matrix.

Proof: Induction on n .

Base Case: If A is an 1×1 matrix, then the statement is trivial.

Inductive steps: Suppose the statement is true for any $(k-1) \times (k-1)$ matrix.

Let $A \in \mathbb{C}^{k \times k}$. Then as $f(\lambda) = \det(A - \lambda I)$ can be factored into k linear factors of the form $(\lambda_i - \lambda)$, A has at least one eigenvalue in $\mathbb{C}$.

Let α be an eigenvalue of A such that there exists a nonzero vector $\vec{x}_1 \in \mathbb{C}^k$ and $A\vec{x}_1 = \alpha\vec{x}_1$.

Extend $\vec{x}_1$ into a basis of $\mathbb{C}^k$, denoted by: $\{\vec{x}_1, \vec{x}_2, \dots, \vec{x}_k\}$

Then $A\vec{x}_1 = \alpha\vec{x}_1$ and $A\vec{x}_j = \sum_{i=1}^k a_{ij} \vec{x}_i$

Denote $P_1 = (\vec{x}_1 \ \vec{x}_2 \ \dots \ \vec{x}_k) \in \mathbb{C}^{k \times k}$ then P_1 is invertible.

$$\Rightarrow AP_1 = A(\vec{x}_1 \ \vec{x}_2 \ \dots \ \vec{x}_k) = (A\vec{x}_1 \ A\vec{x}_2 \ \dots \ A\vec{x}_k)$$

$$= (\alpha\vec{x}_1 \ \sum_{i=1}^k a_{i2} \vec{x}_i \ \dots \ \sum_{i=1}^k a_{ik} \vec{x}_i)$$

$$= (\vec{x}_1 \ \vec{x}_2 \ \dots \ \vec{x}_k) \begin{pmatrix} \alpha & a_{12} & \dots & a_{1k} \\ 0 & a_{22} & \dots & a_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & a_{k2} & \dots & a_{kk} \end{pmatrix}$$

Denote $(a_{12} \ \dots \ a_{1k}) = \vec{a}_1 \in \mathbb{C}^{1 \times (k-1)}$ and $A_1 = \begin{pmatrix} a_{22} & \dots & a_{2k} \\ \vdots & \ddots & \vdots \\ a_{k2} & \dots & a_{kk} \end{pmatrix} \in \mathbb{C}^{(k-1) \times (k-1)}$

$$\Rightarrow AP_1 = P_1 \begin{pmatrix} \alpha & \vec{a}_1 \\ \vec{0} & A_1 \end{pmatrix} \Rightarrow A = P_1 \begin{pmatrix} \alpha & \vec{a}_1 \\ \vec{0} & A_1 \end{pmatrix} P_1^{-1} \Leftrightarrow P_1^{-1} A P_1 = \begin{pmatrix} \alpha & \vec{a}_1 \\ \vec{0} & A_1 \end{pmatrix}$$

By inductive hypothesis there exists an invertible matrix $Q_1 \in \mathbb{C}^{(k-1) \times (k-1)}$ and

an upper triangular matrix $U_1 \in \mathbb{C}^{(k-1) \times (k-1)}$ such that $A_1 = Q_1 U_1 Q_1^{-1}$.

Define $P_2 = \begin{pmatrix} 1 & \vec{0}_{1 \times (k-1)} \\ \vec{0}_{(k-1) \times 1} & Q_1 \end{pmatrix}$ Then $P_2^{-1} = \begin{pmatrix} 1 & \vec{0} \\ \vec{0} & Q_1^{-1} \end{pmatrix} \Leftrightarrow U_1 = Q_1^{-1} A_1 Q_1$

$$\begin{aligned} \text{Note that } P_2^{-1} \begin{pmatrix} \alpha & \vec{a}_1 \\ \vec{0} & A_1 \end{pmatrix} P_2 &= \begin{pmatrix} 1 & \vec{0} \\ \vec{0} & Q_1^{-1} \end{pmatrix} \begin{pmatrix} \alpha & \vec{a}_1 \\ \vec{0} & A_1 \end{pmatrix} \begin{pmatrix} 1 & \vec{0} \\ \vec{0} & Q_1 \end{pmatrix} \\ &= \begin{pmatrix} \alpha & \vec{a}_1 \\ \vec{0} & Q_1^{-1} A_1 Q_1 \end{pmatrix} \begin{pmatrix} 1 & \vec{0} \\ \vec{0} & Q_1 \end{pmatrix} = \begin{pmatrix} \alpha & \vec{a}_1 Q_1 \\ \vec{0} & Q_1^{-1} A_1 Q_1 \end{pmatrix} \end{aligned}$$

$$= \begin{pmatrix} \alpha & \vec{a}, Q_1 \\ 0 & U_1 \end{pmatrix}$$

$$\Rightarrow P_2^{-1} (P_1^{-1} A P_1) P_2 = \begin{pmatrix} \alpha & \vec{a}, Q_1 \\ 0 & U_1 \end{pmatrix}$$

Let $P = P_2 P_1 \Rightarrow PAP^{-1} = \begin{pmatrix} \alpha & \vec{a}, Q_1 \\ 0 & U_1 \end{pmatrix}$ is an upper triangular matrix.

Corollary: Let V be an n -dimensional vector space over $\mathbb{C}$.

Let $T: V \rightarrow V$ be a linear transformation. Then there exists a basis B of V , such that $[T]_B$ is an upper triangular matrix.

Proof. Let $B \subseteq V$ be any ordered basis for V .

Then there exists an invertible matrix P such that $P^{-1}[T]_B P = U$ is an upper triangular matrix.

Denote $P = (P_{ij}) = (\vec{P}_1 \vec{P}_2 \dots \vec{P}_n)$ and denote $B' = \{\vec{v}_1', \vec{v}_2', \dots, \vec{v}_n'\}$

take $\vec{v}_j = \sum_{i=1}^n P_{ij} \vec{v}_i'$ for all $j=1, \dots, n$. i.e. $[\vec{v}_j]_{B'} = \vec{P}_j$

Then $\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\}$ is another ordered basis for V .

(As suppose $a_1 \vec{v}_1 + \dots + a_n \vec{v}_n = \vec{0}$

$$\Leftrightarrow [a_1 \vec{v}_1 + \dots + a_n \vec{v}_n]_{B'} = \vec{0}$$

$$\Leftrightarrow a_1 [\vec{v}_1]_{B'} + \dots + a_n [\vec{v}_n]_{B'} = \vec{0}$$

$$\Leftrightarrow \begin{pmatrix} [\vec{v}_1]_{B'} & \dots & [\vec{v}_n]_{B'} \end{pmatrix} \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} = \vec{0}$$

$$\Leftrightarrow (\vec{P}_1 \dots \vec{P}_n) \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} = \vec{0}$$

$$\Leftrightarrow P \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} = \vec{0} \Leftrightarrow \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} = P^{-1} \vec{0} = \vec{0}.$$

(Review this first).

Then by our result about matrix representation under change of

basis, we have: $[T]_B = P [T]_{B'} P^{-1} = U$

$\Rightarrow [T]_B$ is an upper triangular matrix.

Notation: Let $g(x) = a_0 + a_1x + \dots + a_mx^m \in \mathbb{P}(F)$. Then for any $\forall A \in F^{n \times n}$, then

$$g(A) = a_0I + a_1A + \dots + a_mA^m.$$

Remark: For any $g(x) = (x-a)(x-b)$ and any $A \in F^{n \times n}$

$$(A-aI)(A-bI) = (A-bI)(A-aI).$$

Theorem (Cayley-Hamilton) Let $A \in F^{n \times n}$ ^{where $F = \mathbb{C}$ or $\mathbb{R}$.} and let $f(\lambda)$ be the characteristic polynomial of A . Then $f(A) = 0$.

Proof: For any $A \in F^{n \times n} \subseteq \mathbb{C}^{n \times n}$, there exists an upper triangular matrix

$$U = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix} \in \mathbb{C}^{n \times n} \text{ and an invertible matrix } P \in \mathbb{C}^{n \times n} \text{ such}$$

$$\text{that } A = PUP^{-1}.$$

$$\text{Thus } 1). f(\lambda) = (\lambda - \lambda_1)(\lambda - \lambda_2) \dots (\lambda - \lambda_n)$$

2). Denote $P = (\vec{p}_1 \vec{p}_2 \dots \vec{p}_n)$. Then $\{\vec{p}_1, \dots, \vec{p}_n\}$ forms a basis for $\mathbb{C}^n$.

// end of Febr 2